Predicting the inclination of the Gubbio wall from the legacy sensor set

Station 02, 26 July 2018 to 20 February 2025, hourly grid

Study report · `studies/inclination_prediction_legacy/`

12 August 2026

Abstract

The 14-column network ran at station 02 for six and a half years before the February 2025 changeover. It carries air temperature, relative humidity, battery voltage and inclination — no solar radiation, no wall temperature.

This report asks two families of question. The first is what the extra length buys, and is specific to this era: which periodicities are present, over which stretch they should be decomposed, and how missing values should be filled. The second is the set the extended-sensor study asked — best single predictor, best combination and how to diagnose it, how far ahead and with what uncertainty — repeated here with that study's machinery unchanged, so that any difference in result is a difference in the data and not in the method.

The periodicities are real and identified without imputing anything first: an annual term at 370 d, a semi-annual term at 183 d, and a diurnal term at 23.93 h with its second and third harmonics. A window-function control shows the apparent 971-day feature to be an artefact of the outage pattern. **The semi-annual term is the annual second harmonic**, not an independent 180-day process: frequency alone can never separate the two on a record of this length, but they are phase locked to within 28.5°. **The components do not hold still** — the annual amplitude ranges from 30.6 to 53.3 mdeg — and that is what makes the seasonal model useless for prediction. **Almost all of the forecast skill is autoregressive**: one hour ahead, 47.7 of the 48.8 mdeg climatological error is removed by the signal's own recent history, the seasonal terms remove 0.1, and the drivers remove nothing.

The ported questions answer sharply and negatively. Air temperature is the best single predictor on the whole era at $R^2 = 0.073$; on the unbroken block *no driver beats a constant at all.* The two-driver combination improves on air temperature by 1.3 % against a fold spread twenty-three times that size, and out of period both models roughly double their error. Two warnings for the framework follow: extrapolating a fitted linear trend costs **85–93 mdeg at every horizon**, and only about a tenth of the missing hours can be defensibly filled.

A seventh question was raised by those answers and is addressed in Section 10: how much of the record can be used at all. Everything above was computed under an assumption stated in Section 2 and never tested — that autoregressive work must run on one unbroken block. It need not. An autoregressive model consumes windows, not a continuous series, and fitting across every complete stretch of the era yields **6.57 annual cycles and 2.26 times the training windows for 1.5 % fabricated data**. The consequence is large: the deployable configuration's prediction intervals fall from 164–185 mdeg, which excluded nothing, to **12–23 mdeg**, and its one-hour error by 44 %, with no change to the model. The same comparison tests a rule the companion framework asserts — that one to two annual cycles suffice for a yearly component — and finds it half right: the longer span corrects the annual amplitude but lets NeuralProphet's trend absorb *more* of the low-frequency variance, 53.8 % against 39.9.

Contents

1 What this study asks, and what it assumes

1.1 Six questions, three of them ported

Specific to the long record.

1. Which periodicities are present in the compensated inclination?
2. Over which stretch should they be decomposed, and do they persist?
3. How should missing values be filled, and up to what gap length is that defensible?

Ported from `studies/inclination_prediction/`, for comparability.

4. Can one sensor channel predict the inclination, and which is best?
5. Can a combination beat the best single channel, and how is the best combination diagnosed?
6. How far ahead can NeuralProphet forecast, with what uncertainty — and what is the error at each horizon actually made of?

Question 6 merges the sibling study's horizon question with this study's error budget. That is the natural place for it: a horizon curve says how large the error is, and the budget says which component produced it.

1.2 Premises, imposed rather than tested

Premise	Consequence for this report
The logged `I` channel is an inclination in millidegrees	The 2500 offset is added back. In this era the logged values run 2105–2300, so read as millidegrees they lie **outside** the ±2000 mdeg the datasheet certifies. Accepted by instruction; the compensation normalises the series to start at zero, so nothing downstream depends on it.
The documented compensation is correct	`inc_comp` is the target of every model. Section 9 shows this is by far the most consequential of the three.
The analysis grid is one hour	Matching every external proxy the pipeline aligns against.

1.3 The sign contradiction, reproduced on a second instrument

The site predicts a negative association between the inclination and any heating driver: the valley face is the more sun-exposed, expands further, and tips the wall towards the mountain. As in the extended-sensor study, that prediction is tested on the *raw* channel, before the compensation can manufacture the expected sign.

Series	Driver	r lev.	slope lev.	r band	slope band
Raw `inc`	`tair`	+0.241	+1.173	+0.957	+2.379
	`rh`	−0.244	−0.479	−0.584	−0.343
`inc_comp`	`tair`	−0.629	−3.827	−0.964	−2.621
	`rh`	+0.301	+0.738	+0.595	+0.381

Slopes in mdeg per °C for temperature and per percentage point for humidity; "band" is the diurnal band.

The same positive slope on a different instrument

The legacy inclinometer moves *positively* with air temperature, at +2.379 mdeg/°C on the diurnal band. The extended-package instrument, installed seven years later as a physical reinstallation sharing no baseline and no hardware with this one, measured +2.291.

Two independent instruments, agreeing to within 4 % on a slope that runs opposite to the site's structural expectation. That is much harder to explain as a per-instrument defect than as a systematic property of the instrument type or of the sign convention itself, and it correspondingly weakens the thermal-drift reading of `docs/raw-data-format.md` Section 7.5 in favour of an inverted convention.

This is new evidence and it does not settle the question, which still needs the calibration sheet or the mounting orientation. It does mean the contradiction is a property of the measurement chain rather than of one sensor.

2 The record, and what it can support

2.1 Loading and cleaning

Of the 1926 legacy files, a third carry duplicate timestamps whose copies disagree, an eighth mix decimal separators internally, and some carry records dated to the previous day. Duplicates are merged field by field rather than dropped, because where copies differ it is usually because one carries real values where the other carries zeros: 19 210 duplicated records were merged, resolving 18 473 field-level conflicts. Sentinels sent 64 248 zeros to `NaN`. The spike screen removed nothing — unlike the current era, this record has no acquisition-restart transient.

The result is 57 625 hourly slots of which 39 499 carry the inclinometer, that is **68.5 % coverage**.

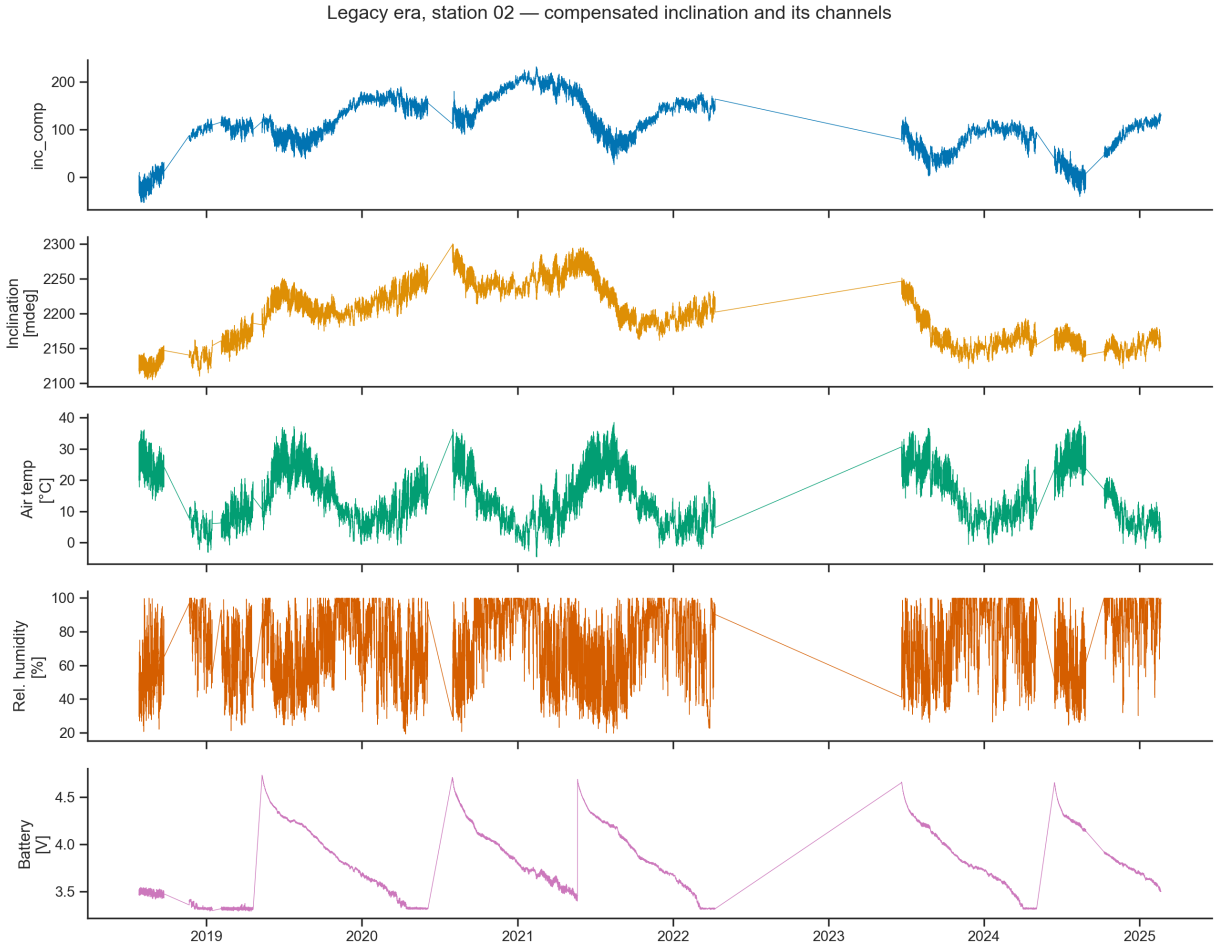

Figure 1: The legacy era in full. The compensated inclination spans roughly 285 mdeg against a diurnal amplitude of tens, and the outages are visible as the blank stretches — five of them longer than thirty days.

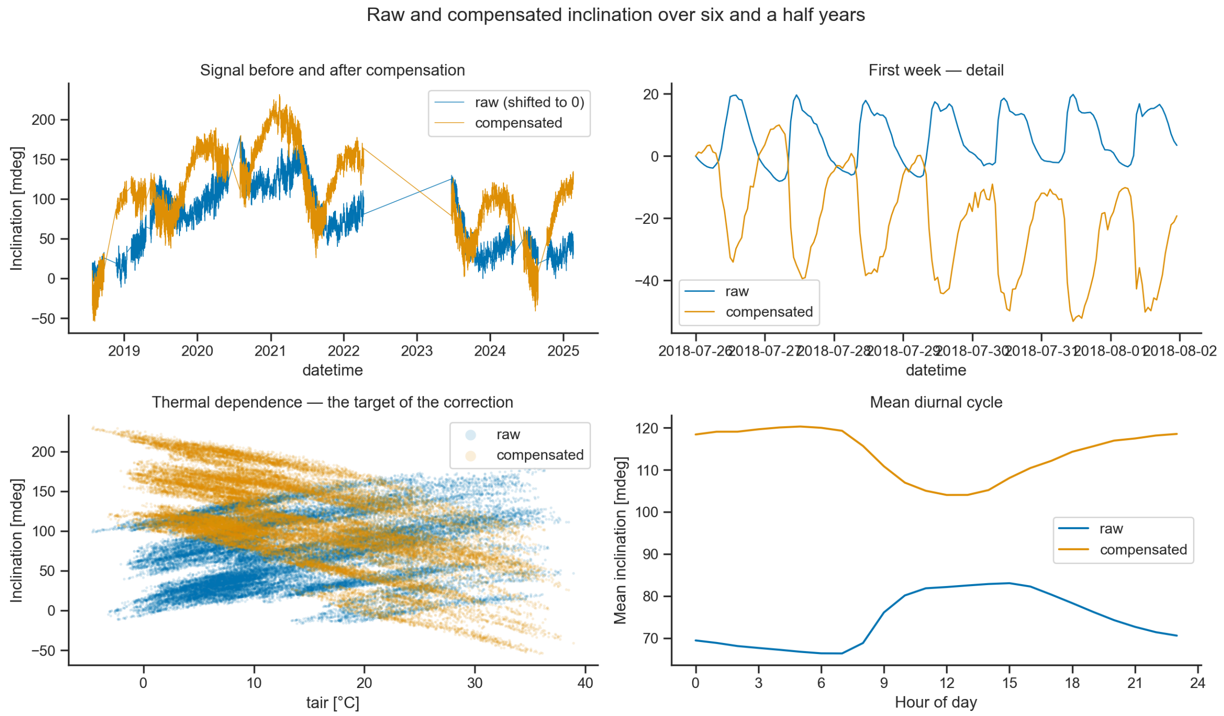

Figure 2: Raw and compensated inclination with the correction that separates them. As in the current era, the correction is not a small adjustment.

2.2 Where the missing hours are

Gap length	gaps	hours lost	share of missing
$\leq$ 3 h	417	606	3.3 %
3–6 h	58	275	1.5 %
6–24 h	70	875	4.8 %
1–3 d	2	65	0.4 %
3–7 d	0	0	0.0 %
7–30 d	2	989	5.5 %
> 30 d	5	15 316	**84.5 %**

The shape of this table decides the imputation question before any method is tried. There are 554 gaps, but 84.5 % of the lost hours sit in five of them.

2.3 No block contains two annual cycles

Longest blocks, 72 h tolerance	start	days	coverage	cycles
first	2020-07-31	617.2	96.8 %	**1.69**
second	2019-05-11	390.1	97.7 %	1.07
third	2023-06-21	317.5	91.8 %	0.87
fourth	2024-10-09	133.4	99.3 %	0.37

This divides the study in two

Even absorbing interruptions of up to three days, the longest unbroken stretch holds **1.69** annual cycles. Nothing in this record qualifies on a two-cycle rule.

The annual term therefore cannot be identified from any single contiguous window. It has to be fitted *across* the gaps — which a least-squares harmonic fit does naturally and an autoregressive model cannot. So the spectrum and the harmonic decomposition run on the **whole era**; the autoregressive work runs on the **longest block**, 2020-07-31 to 2022-04-09; and the third block, 2023-06-21 to 2024-05-04, is held back entirely for the out-of-period test of Section 7.4.

3 Question 1 — the spectrum

3.1 Why not a periodogram

A fifth of the record is missing and the missingness is structured into a handful of long outages. An ordinary periodogram needs a complete grid, and filling those holes before estimating the spectrum lets the filling method decide the answer: a 271-day hole interpolated linearly injects power at exactly the low frequencies under investigation.

The Lomb–Scargle periodogram fits a sinusoid at each trial frequency by least squares over whatever samples exist, so it consumes the gaps rather than requiring them filled. **The spectrum is computed before any imputation.**

Two controls accompany it. The **window function** is the same periodogram run on the observation mask alone: any peak appearing there is a property of when the instrument was running. The **permutation null** shuffles values against observation times, destroying periodicity while preserving the sampling pattern exactly; two hundred replicates put the 95th and 99th percentiles at 0.0124 and 0.0140.

3.2 What is there

Period	power	window	above 99 %	reading
370.1 d (1.01 yr)	**0.3995**	0.0279	yes	annual, genuine
971.0 d (2.66 yr)	0.1178	**0.1336**	yes	**fail** outage artefact
544.6 d (1.49 yr)	0.0538	0.0207	yes	no physical reading
183.0 d (0.50 yr)	0.0351	0.0211	yes	semi-annual, genuine
113.9 d (0.31 yr)	0.0278	0.0081	yes	unanticipated, genuine
75.2 d	0.0052	0.0027	no	below the null

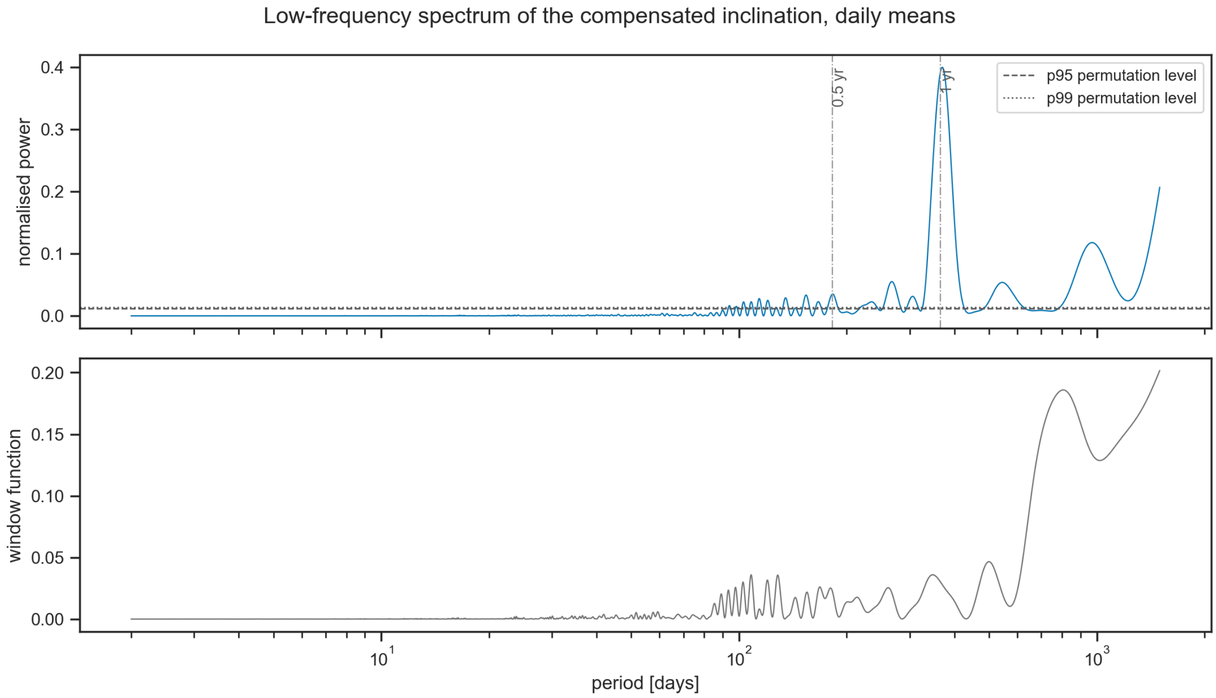

Figure 3: Low-frequency Lomb–Scargle spectrum on daily means, with the permutation levels, and the window function beneath. The rise in the window function beyond about 700 days is what disqualifies the 971-day feature.

The window-function control earns its place

The second-largest peak in the data spectrum, at 971 days, has a *window-function power larger than its own* — 0.1336 against 0.1178. It is produced by the arrangement of the outages, not by the wall. Without this control it would have been reported as a 2.7-year structural cycle, comfortably clearing the permutation null.

The three periodicities anticipated before the analysis — one day, roughly six months, one year — are all present and all clear the null. A fourth, at **113.9 days**, was not anticipated. It is well separated from the window function, so it is not an outage artefact, and it has no obvious harmonic relationship to the annual term. This report records it and does not explain it.

3.3 The diurnal band

Period	power	reading
23.93 h	**0.0407**	diurnal
7.28 d	0.0068	**fail** artefact of the 168 h detrending filter
11.99 h	0.0017	second harmonic
8.00 h	0.0015	third harmonic

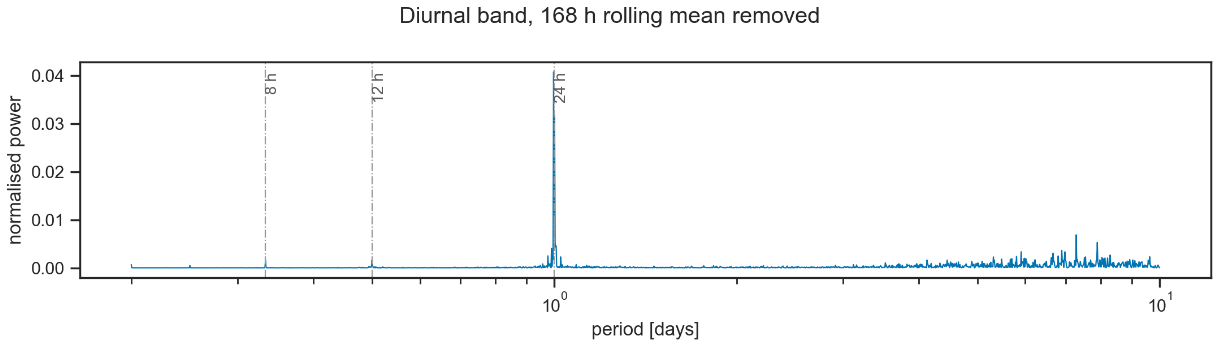

Figure 4: The diurnal band after removing a one-week rolling mean. The daily cycle and its second and third harmonics are clear.

The 7.28-day feature is a property of the analysis. A rolling mean of width W acts as a high-pass whose response peaks near W itself, and 168 h is seven days; re-running with a 720-hour window moves the feature, which is the test that identifies it. A Welch periodogram on the longest unbroken block confirms the daily peak independently at 1.0000 d.

4 Question 2 — decomposition, and whether it holds still

4.1 One limit that no amount of analysis removes

Separating an independent **180.0-day** process from the annual second harmonic at **182.62 days** requires resolving a frequency difference of 8.0×10^{-5} cycles per day. That needs a record of roughly **34 years**. With six and a half it cannot be done, and no choice of estimator, window or grid changes that.

Phase behaviour separates them where frequency cannot. If the semi-annual term is harmonic distortion of the annual cycle — which any nonlinear response to a sinusoidal annual driver produces automatically — its phase advances at exactly twice the annual rate, so $2\varphi_{\text{annual}} - \varphi_{\text{semiannual}}$ stays constant. An independent process drifts.

4.2 Amplitudes, phases, and their drift

Component	period	amplitude	s.e.	peak, days after start	var. share
annual	365.25 d	**44.56**	1.36	208.6	0.376
semiannual	182.62 d	13.99	1.34	112.8	0.037
diurnal	1.00 d	8.04	0.20	0.1	0.012
semidiurnal	0.50 d	2.96	0.10	0.3	0.002

Amplitudes in mdeg; Newey–West standard errors with 24 lags. Model $R^2 = 0.487$.

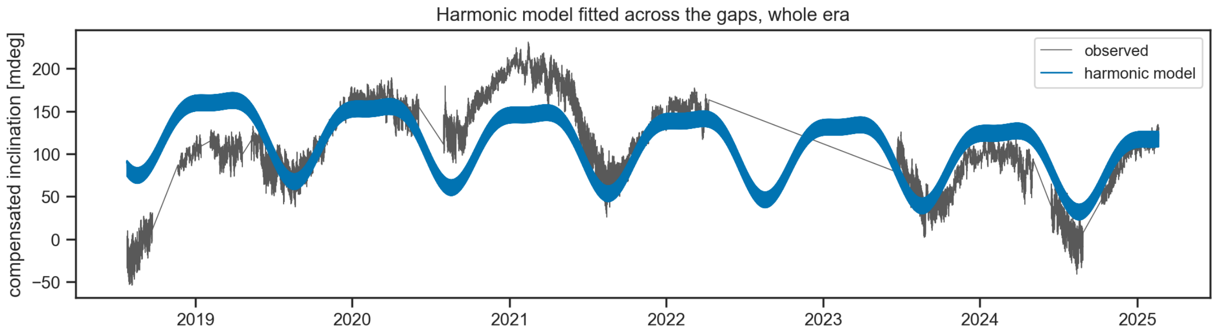

Figure 5: The harmonic model fitted across the gaps over the whole era. It passes through the outages because a least-squares harmonic model does not require continuity.

Refitting in two-year windows stepped by ninety days:

Component	mean amp.	s.d.	amplitude range	phase range
annual	41.89	8.72	30.6 – 53.3	189.5 – 215.0 d
semiannual	9.14	3.07	4.7 – 13.9	87.7 – 134.5 d

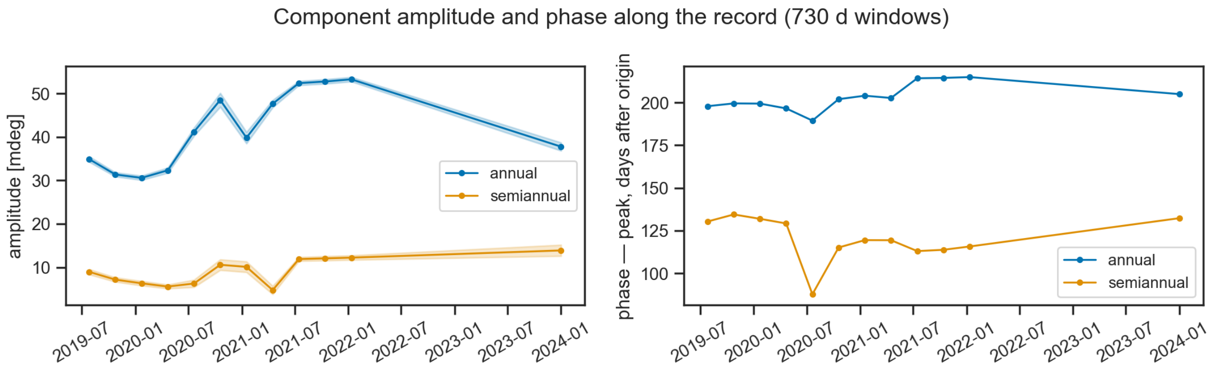

Figure 6: Amplitude and phase of the annual and semi-annual components along the record. Neither is constant: the annual amplitude varies by a factor of 1.7 and its phase by 25 days. This figure explains why the seasonal model fails to predict in Section 8.

The semi-annual term is the annual second harmonic

The phase-lock test over twelve windows gives a circular standard deviation of $2\varphi_{\text{annual}} - \varphi_{\text{semiannual}}$ of **28.5°**, resultant length 0.884, against **103.9°** for an unlocked null. The two components track each other.

The 183-day peak is therefore harmonic distortion of a non-sinusoidal annual cycle, not a separate process with a six-month period. Its amplitude carries information about the *shape* of the annual response, not about any independent semi-annual forcing.

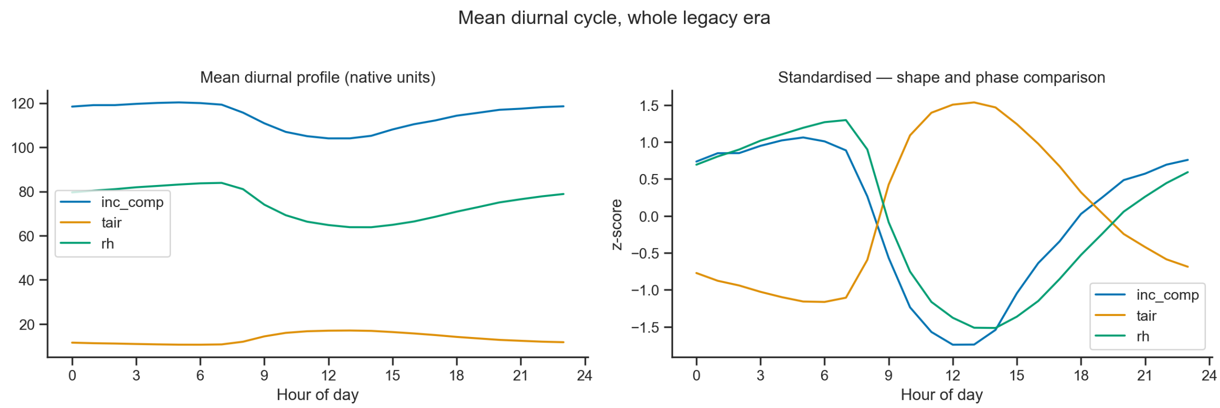

Figure 7: Mean diurnal cycle over the whole era, for the target and both drivers.

5 Question 3 — imputation

Filling a hole is defensible only up to the length at which the filler stops knowing anything the data did not already contain. Forty randomly placed gaps per length, eight lengths, four fillers.

gap	interpolate	seasonal naive	harmonic	ridge, drivers
1 h	**1.12**	6.14	37.80	42.00
3 h	**1.59**	4.51	28.34	32.32
6 h	**3.18**	5.19	27.77	34.51
12 h	6.07	**5.37**	25.44	34.99
24 h	7.11	**4.40**	33.44	37.69
72 h	8.15	**6.90**	27.78	37.03
168 h	8.73	**8.27**	25.71	32.74
720 h	**8.23**	22.93	22.44	32.17

MAE in mdeg, against a signal standard deviation of 51.6 mdeg.

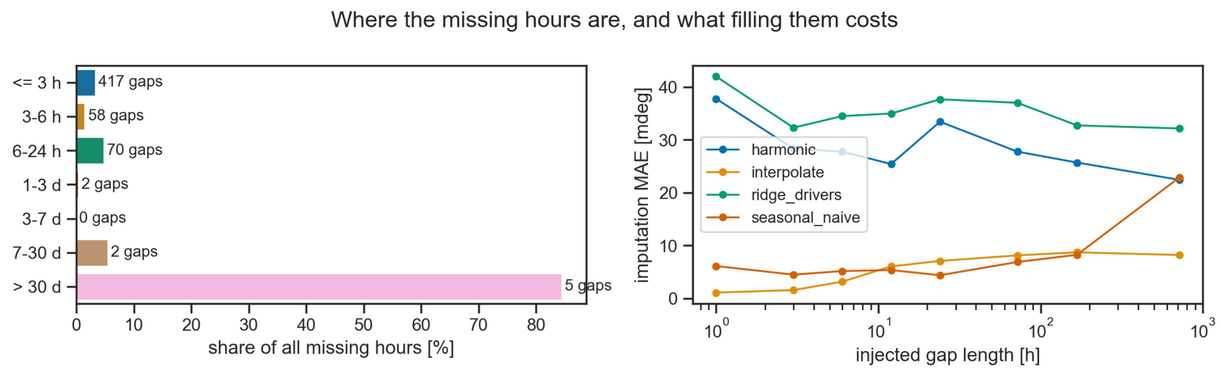

Figure 8: Left: where the missing hours are, by gap length. Right: what each filler costs against injected gaps of known length.

Applying a five-millidegree tolerance — a tenth of the signal's standard deviation — gives **interpolation to 6 h** and **seasonal-naive to 24 h**; the model-based fillers never qualify.

Two results deserve emphasis. The **model-based fillers are the worst of the four at every gap length**, by a wide margin: the harmonic model explains under half the variance and knows nothing about local behaviour, and the driver-based ridge is worse still — which anticipates

Section 8's finding that these drivers carry almost no predictive content. Simple interpolation beats both by a factor of eight at short gaps.

And the policy interacts with the gap inventory to give the practical answer: filling everything up to 24 hours addresses gaps holding **9.6 %** of the missing hours. The 84.5 % locked in stretches longer than thirty days is not recoverable by any method tested, and should be treated as absent rather than reconstructed.

6 Question 4 — operators and the best single predictor

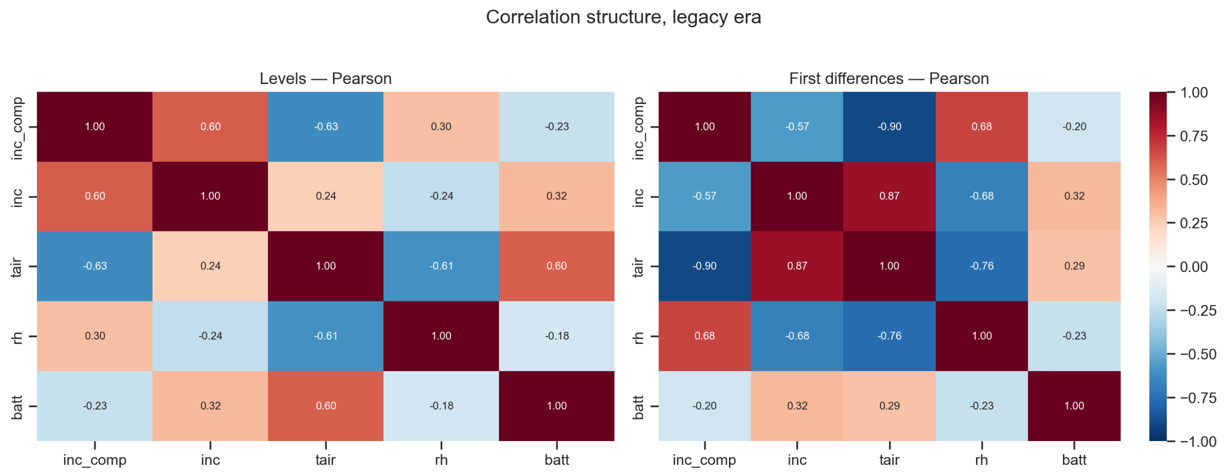

Figure 9: Correlation structure of the legacy channels, on levels (left) and on first differences (right). Levels correlations between two trending series are inflated by the shared trend, so the differenced matrix is the honest one for a signal with drift. Note that the raw and compensated inclination sit at opposite signs against air temperature, which is Section 1.3 seen from a second angle.

6.1 Full band against diurnal band

Before any driver is ranked it is given the chance to act through the operator that suits it: a transport delay, and a thermal inertia. The delay is bounded at twelve hours because both candidates are **external forcings**, which must causally precede the response (`docs/raw-data-format.md` Section 7.5).

Driver	**delay**	τ	r	R^2	R^2 **at** $d = \tau = 0$
Full band					
`tair`	0 h	0 h	−0.629	0.395	0.395
`rh`	0 h	168 h	+0.382	0.146	0.091
Diurnal band — carried forward					
`tair`	0 h	0 h	−0.964	**0.929**	0.929
`rh`	0 h	0 h	+0.595	0.353	0.353

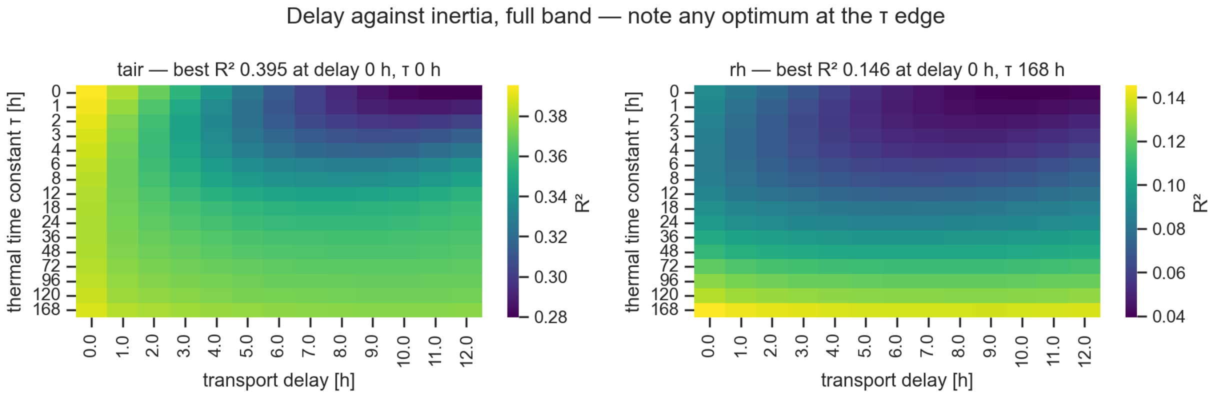

Figure 10: Delay against inertia, full band. Relative humidity's optimum sits at $\tau = 168\,\mathrm{h}$, the last value in the grid.

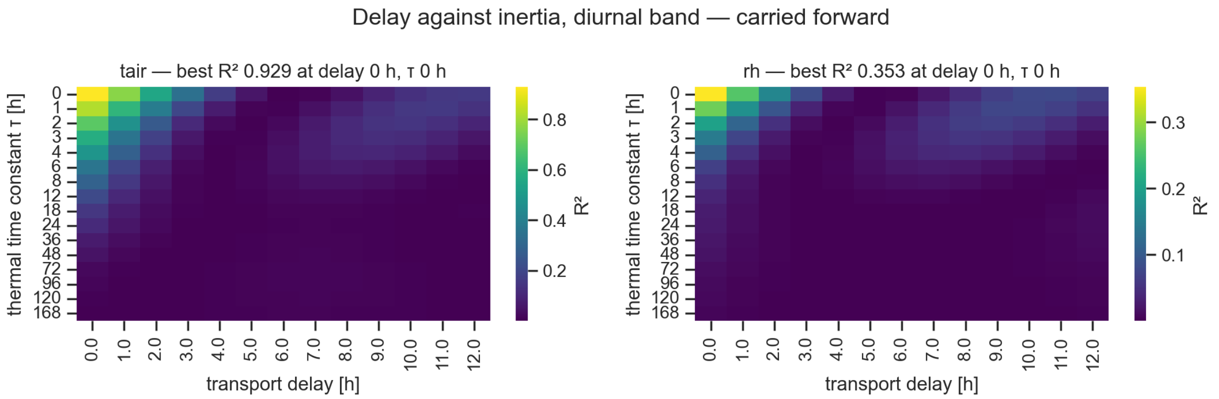

Figure 11: The same grid on the diurnal band. Both optima collapse to zero delay and zero time constant.

The comparison reproduces the extended-sensor study's finding exactly. On the full band, relative humidity's optimum runs to the *edge* of the τ grid, lifting explained variance from 0.091 to 0.146. A time constant of one week is not a thermal property of a masonry wall; it is the width of filter needed to turn a driver into a seasonal envelope, and it is pinned at the boundary because a longer filter would fit better still. Band-limiting removes it. Reporting the full-band operator as a physical time constant would have produced a confident and spurious finding.

6.2 The signed-delay control

The extended-sensor study had to scan wall temperature over *signed* delays, because an internal state variable at unknown depth may legitimately lag the deformation. This era has no wall probe: both candidates are external forcings, so the signed scan here is a **control rather than a measurement**. If either driver gained materially from a lead, the method would be at fault.

Driver	delay	τ	R^2 signed	R^2 causal	gained from lead
`tair`	0 h	0 h	0.9291	0.9291	0.0000
`rh`	−1 h	0 h	0.3552	0.3534	0.0018

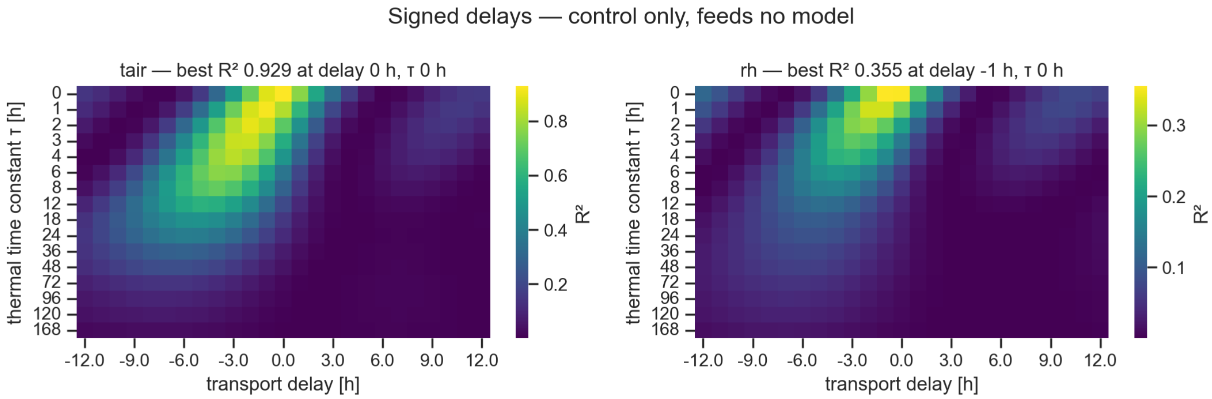

Figure 12: The diurnal-band grid over signed delays. Control only; feeds no model.

The control passes. Air temperature peaks exactly at zero. Relative humidity peaks one hour early, but that lead buys **0.0018** of explained variance — an order of magnitude below the 0.01 threshold the study sets for a material lead, and exactly what an optimum landing one grid step off zero on noise looks like. Neither forcing leads the response in any meaningful sense, which is what the physics requires.

6.3 Ranking, on two windows

Each driver is scored on its own, at its diurnal-band operator, by pooled error over five rolling-origin folds. Autoregression is excluded: with it, everything ties behind the autoregressive term and the ranking says nothing. Battery voltage is carried as a **negative control**.

Driver	**MAE**	**fold sd**	R^2	**vs best**
Whole era				
`tair`	36.70	11.19	+0.073	—
`batt` †	38.89	11.96	−0.215	+6.0 %
`rh`	41.29	13.13	−0.327	+12.5 %
Unbroken block only				
`tair`	38.04	16.19	−0.719	—
`rh`	41.83	23.51	−1.403	+10.0 %
`batt` †	46.05	19.43	−1.583	+21.1 %

† negative control.

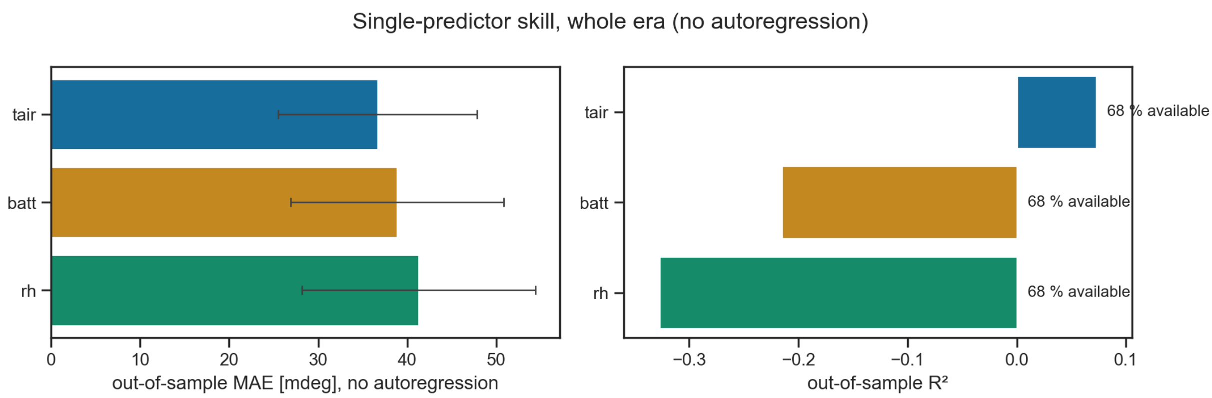

Figure 13: Single-predictor skill on the whole era, without autoregression. Only air temperature beats a constant, and only barely.

On the unbroken block, no driver beats a constant

Air temperature's R^2 falls from +0.073 on the whole era to −**0.719** on the longest unbroken block — worse than predicting the mean. The negative control now ranks last, which is the one reassuring feature of the table.

The marginal skill air temperature shows on the full era does not survive a change of window. Whatever it was tracking was a property of the particular mix of periods the whole record contains, not a stable relationship. This is the clearest statement in the report that **the legacy drivers do not predict this signal**, and Sections 8 and 9 reach it independently.

One discrepancy to read carefully

Air temperature explains **92.9 %** of the variance of the *diurnal band* ($r = -0.964$) and **7.3 %** of the total variance out of sample. Both numbers are correct and they are not in conflict: the diurnal band is a small share of this signal's variance, which is dominated by seasonal and lower-frequency structure an instantaneous driver cannot reach.

Quoting the band-limited correlation as though it described the signal would overstate what this channel knows by an order of magnitude.

7 Question 5 — combinations, and how to diagnose them

7.1 Collinearity

Channel	VIF	max $\|r\|$ with others	closest
`tair`	1.58	0.605	`rh`
`rh`	1.58	0.605	`tair`

This is a useful contrast with the extended era, where air and wall temperature were near-duplicates at $r = 0.924$ with variance inflation factors around eight. Here the two candidates are only moderately related, so the collinearity caveat that governed the sibling study's diagnostics does not bite: coefficients would be readable. They are still not read, because the out-of-sample error is the quantity that matters.

7.2 Every combination

With two candidates the exhaustive search is three subsets.

Subset	k	MAE	fold sd	R^2	BIC	utility
`tair + rh`	2	**36.25**	10.83	+0.095	403 197	**0.685**
`tair`	1	36.70	11.19	+0.073	403 836	0.676
`rh`	1	41.29	13.13	−0.327	419 943	0.601

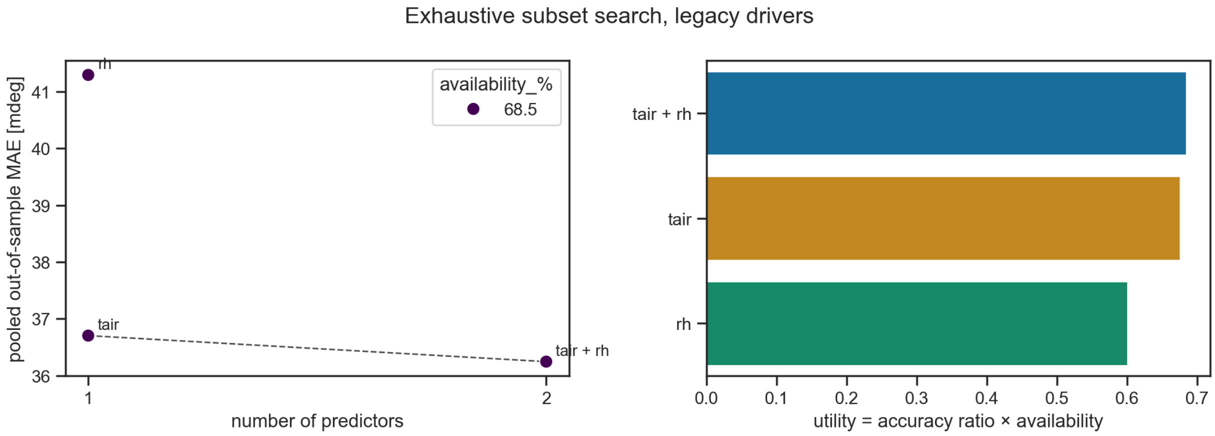

Figure 14: Exhaustive subset search over the two legacy drivers.

The pair beats air temperature alone by **0.46 mdeg, that is 1.3 %**, against a fold-to-fold spread of 10.8 mdeg — twenty-three times the difference. **The ordering is not resolved by these data.** BIC and utility agree with the error ranking, but all three are reading the same unresolved margin.

7.3 Which member is doing the work

Channel	MAE without	gain	permutation increase	sd
`tair`	41.29	5.05 (12.2 %)	**9.07**	2.69
`rh`	36.70	0.46 (1.3 %)	−0.006	0.124

Subset `tair + rh`; full-subset MAE 36.25 mdeg. All values in mdeg.

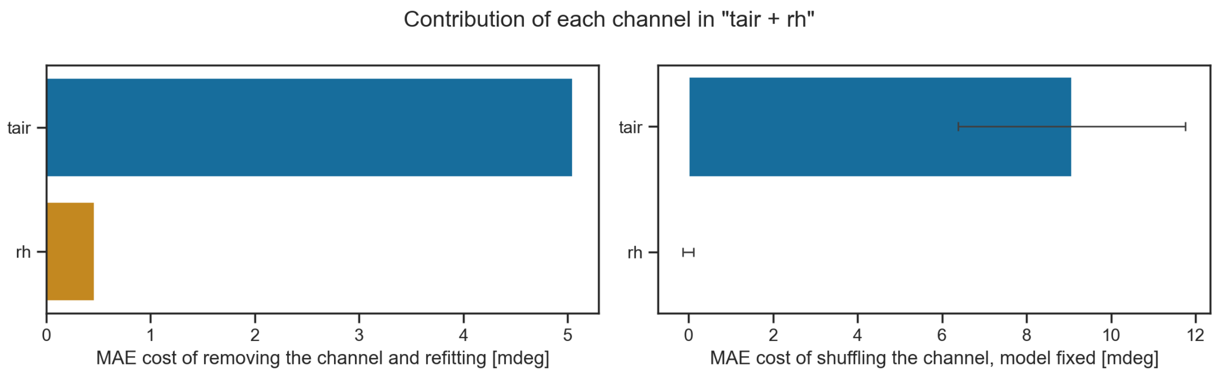

Figure 15: Left: the cost of removing each channel and refitting. Right: the cost of shuffling it with the model held fixed. Relative humidity sits at zero on the right-hand panel, inside its own error bar.

The two diagnostics agree here, which is itself informative. Relative humidity's leave-one-out gain is 0.46 mdeg and its permutation increase is -0.006 ± 0.124 — indistinguishable from zero. The fitted model does not rely on it, and refitting without it costs nothing separable from fold noise. Under the low VIF above there is no collinearity to hide behind: relative humidity simply carries no content.

7.4 The out-of-period test

Both models are fitted on the unbroken block and applied to the third block — 2023-06-21 to 2024-05-04, on the far side of a 271-day outage — without refitting.

Model	n	MAE	RMSE	vs cross-validated
`tair + rh`	6997	**62.49**	65.00	×1.72
`tair` alone	6997	63.56	65.92	×1.73

Neither model transfers

Both roughly **double** their cross-validated error, from 36 to 62–64 mdeg — worse than the 48.8 mdeg a constant achieves in the error budget of Section 8. Across a 271-day outage, a model fitted on environmental drivers does not merely degrade; it becomes worse than predicting the mean.

The pair is marginally better than the single predictor here, reversing the extended-sensor study's holdout result — but by 1.07 mdeg on two models that are both failing, which settles nothing.

7.5 The diagnostic protocol

The procedure used above is the answer to "how is the best combination diagnosed". It is the sibling study's protocol with one rule added — rule 2 — which this era's spectral work forced.

	Rule	Failure it prevents
1	Bound the operator scan by the mechanism and by half the dominant period	A physically impossible optimum, or an aliased lead read as a delay
2	Estimate the spectrum before imputing anything, and control for the sampling pattern	Letting the filling method decide which frequencies exist; reading an outage pattern as a structural cycle
3	Fit each driver operator on the band of interest	A boundary optimum reported as a physical time constant
4	Score without autoregression	Every driver tying behind the autoregressive term
5	Rolling-origin folds, report the spread	A ranking that depends on where the single split fell
6	Exhaustive subset search when the candidate set is small	Selection bias and instability under collinearity
7	Gain and permutation importance, never coefficients under high VIF	Importance attributed by collinearity rather than content
8	Weight by availability over the whole record	Choosing a channel absent when it is needed
9	Carry a negative control	Mistaking shared diurnal shape for predictive content
10	Confirm on an out-of-period block	A combination that fits one period and transfers to none

On this record rules 5, 9 and 10 did the work: the fold spread showed the subset ordering to be unresolved, the negative control ranked above relative humidity on the whole era, and the out-of-period test showed both models failing.

8 Question 6 — horizon, uncertainty and the error budget

8.1 The ladder

The ladder adds one component at a time and scores every rung at every horizon, so each component's contribution is read off consecutive rows. Every rung is fitted before a chronological cut and scored across the whole test region at an exact lead time, pooled over three origins.

Rung	1 h	24 h	168 h	720 h	2160 h	4380 h	8760 h
climatology	48.80	48.82	48.98	49.56	50.57	51.07	**51.67**
+ annual	50.12	50.15	50.32	50.96	52.42	55.12	55.43
+ semiannual	49.44	49.46	49.56	50.01	51.58	54.75	54.98
+ diurnal	49.34	49.36	49.46	49.90	51.47	54.70	54.98
+ AR	**1.60**	**4.90**	**8.35**	**12.27**	**23.49**	**45.03**	61.55
+ drivers	1.60	4.95	8.13	12.61	25.19	46.96	62.87

MAE in mdeg. Horizons 3, 6, 12 and 72 h omitted for width; they interpolate smoothly.

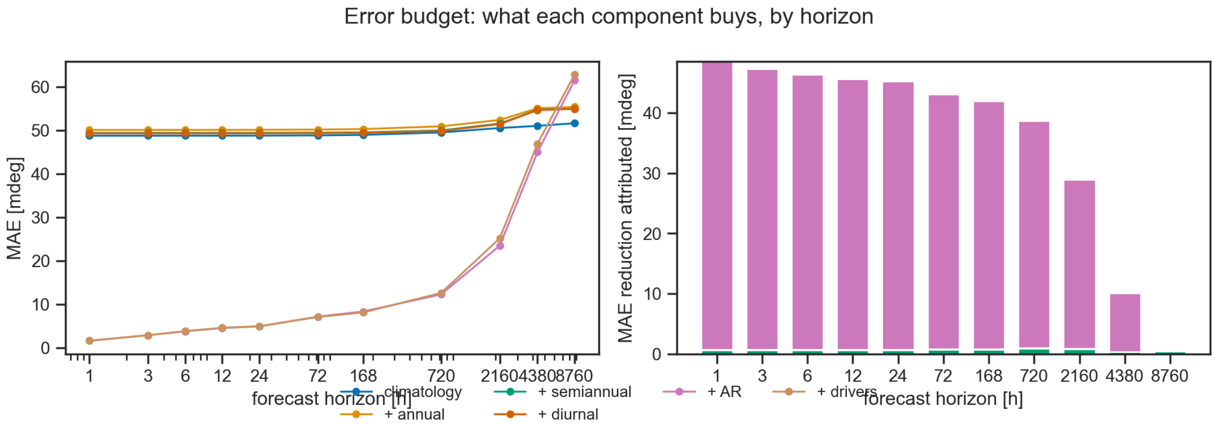

Figure 16: Left: error against horizon per rung. The calendar-only rungs are flat, because a function of time does not degrade with lead; the autoregressive rungs climb. Right: the error reduction attributed to each component.

Autoregression is almost the entire budget. At one hour it removes 47.74 of the 48.80 mdeg climatological error, and it remains dominant out to 720 h — one month — where it still removes 37.63. It continues to beat climatology at 4380 h (45.03 against 51.07) and finally fails at one year.

The seasonal terms buy nothing, and the annual term is actively harmful. Adding it makes the forecast *worse* by 1.33 mdeg at every horizon; the semi-annual term recovers 0.68 and the diurnal term 0.10. The reason is Figure 6: the annual amplitude ranges over a factor of 1.7 and its phase by 25 days, so a fixed harmonic fitted on the training block does not describe the test block. The cycle is real — the spectrum is unambiguous — but it is not stationary, and only a stationary component can be extrapolated.

The drivers add nothing measurable, at every horizon, within ±0.25 mdeg.

8.2 What the trend term costs

Model	1 h	24 h	720 h	4380 h	8760 h
seasonal, no trend	49.34	49.36	49.90	54.70	54.98
seasonal + trend	134.34	134.38	135.65	143.92	148.15
cost	**+85.00**	**+85.02**	**+85.75**	**+89.22**	**+93.18**

Do not extrapolate a fitted linear trend on this signal

A straight line fitted on part of this record and extended across the rest costs between 85 and 93 mdeg at every horizon — against a signal standard deviation of 51.6 mdeg and a climatological error of 49. It is by a wide margin the most damaging single term available, and it is a term the grey-box pipeline includes by default.

The cause is the same non-stationarity that defeats the annual harmonic. A drift term is defensible as a *description* of a fitted window; as an *extrapolation* it is not.

8.3 Knowing the drivers in advance

horizon	lagged	future	gain
1 h	1.60	1.61	−0.5 %
24 h	4.95	4.26	+14.0 %
168 h	8.13	4.99	+38.6 %
720 h	12.61	9.43	+25.2 %
8760 h	62.87	61.82	+1.7 %

Perfect foreknowledge of the drivers is worth nothing at one hour, most at one week, and almost nothing again at one year. At short horizons the signal's own history already contains what the drivers would say; at long horizons neither knows anything.

8.4 NeuralProphet at the short horizons

NeuralProphet cannot span a year, so it is run on the longest unbroken block at horizons to one week.

Correction: the annual term was never enabled

An earlier version of this section stated that this run carried the annual term, and qualified the result by noting that the block holds only 1.69 cycles so the component would be weakly identified. That qualification described something that does not exist. The function producing these numbers, `ip.run_horizon_experiment`, fixed `yearly_seasonality` to false, so the configuration below carries a daily seasonality, an autoregressive term and the drivers, and no annual component whatever.

The numbers in this section are unaffected — they always described the model that was actually fitted. What was wrong was the description. The parameter is now explicit, and Section 10 enables it on a span long enough to identify it.

h	MAE	persist.	skill vs p.	cover.	width
lagged — the deployable configuration					
1	3.35	1.55	−1.160	100.0 %	164.2
12	4.76	6.74	+0.294	100.0 %	165.5
24	5.66	4.41	−0.283	100.0 %	167.0
168	13.53	7.59	−0.783	99.4 %	185.3
future — regressors known over the horizon					
1	2.01	1.54	−0.305	61.5 %	14.5
12	3.15	6.65	+0.526	60.6 %	15.7
24	3.56	4.38	+0.188	61.2 %	16.8
168	6.27	7.43	+0.156	65.0 %	21.9

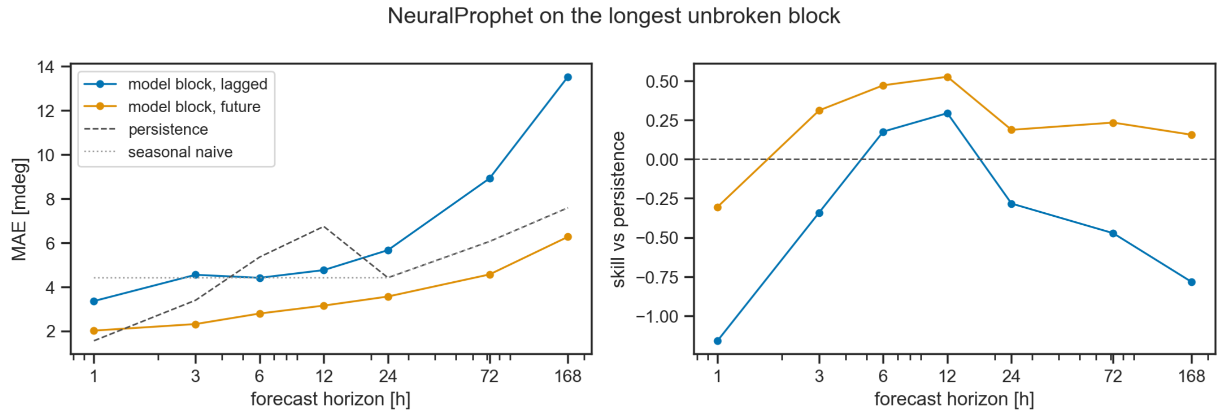

Figure 17: NeuralProphet error and skill against persistence on the longest unbroken block.

The deployable configuration beats persistence only between 6 and 12 hours, and beats the seasonal-naive baseline only out to 6 hours. With perfect future regressors the useful band extends to a full week.

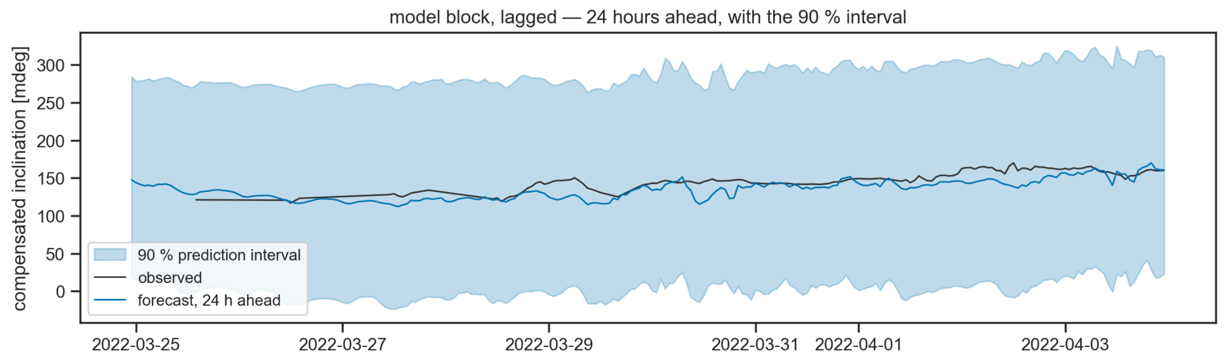

Figure 18: Ten days of the test period at a twenty-four-hour horizon, with the nominal 90 % interval. The interval is so wide that the observed series never approaches its edges.

8.5 Is the quoted uncertainty honest?

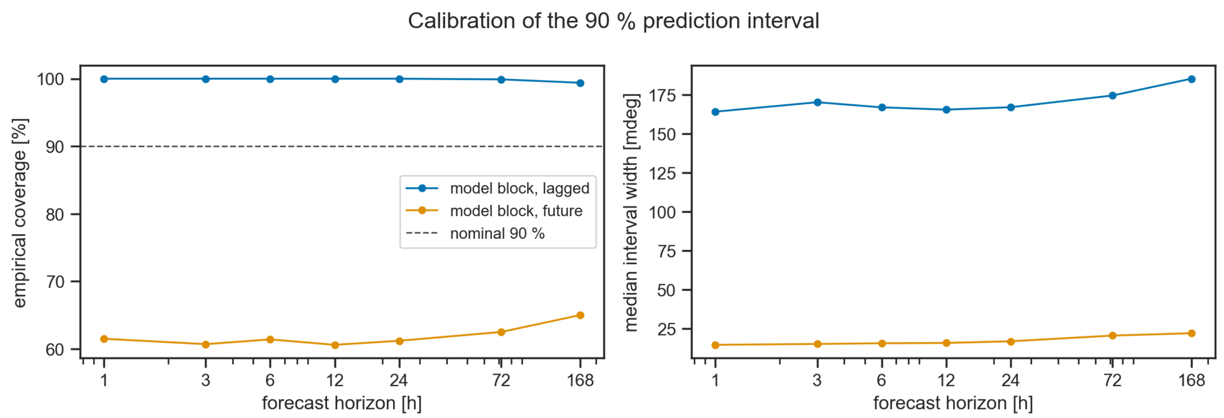

Figure 19: Calibration and width of the nominal 90 % interval. The lagged configuration covers 100 % of observations — by producing intervals more than three times the signal's own standard deviation.

Both configurations fail calibration, in opposite directions

The **lagged** model covers **100 %** of observations against a nominal 90, which is not a success: it achieves this with a median interval width of 164–185 mdeg, more than three times the signal's standard deviation of 51.6. An interval that wide excludes nothing and so detects nothing.

The **future** model covers **61–65 %** with widths of 14–22 mdeg — informative in width, but substantially under-covering.

The extended-sensor study found its deployable configuration under-covering at 54–67 %. Here the same configuration over-covers by inflating the interval. Both are miscalibrated, in opposite directions, which is a reason to recalibrate by a split-conformal procedure rather than to trust either quantile head.

8.6 What the joint multi-horizon head costs

One fit produced every horizon. Refitting with `n_forecasts` set to exactly one horizon isolates the cost of sharing weights across 168 output steps.

h [h]	dedicated	joint head	joint penalty
1	2.93	3.35	+14.3 %
24	5.96	5.66	−5.0 %

The joint head costs 14 % at one hour and *gains* 5 % at twenty-four — the same pattern the extended-sensor study found, consistent with shared weights acting as a regulariser at longer horizons and a distraction at the shortest. The conclusion is unchanged: even the dedicated one-hour model, at 2.93 mdeg, does not beat persistence at 1.55.

9 Sensitivity

The headline is the autoregressive rung's error one day ahead, 4.90 mdeg.

Variation	value	change	change %
baseline	4.90	—	—
compensation coefficient 0.010	13.88	+8.98	+183.2 %
model block only	9.46	+4.56	+93.0 %
compensation coefficient 0.00163	2.07	−2.83	−57.7 %
autoregressive depth 48	4.78	−0.12	−2.4 %
autoregressive depth 12	4.98	+0.08	+1.6 %
origins 0.4 / 0.5 / 0.6	4.92	+0.02	+0.4 %
no seasonal terms at all	4.89	−0.01	−0.2 %
no semi-annual term	4.90	−0.00	−0.1 %
drivers dropped	4.90	0.00	0.0 %

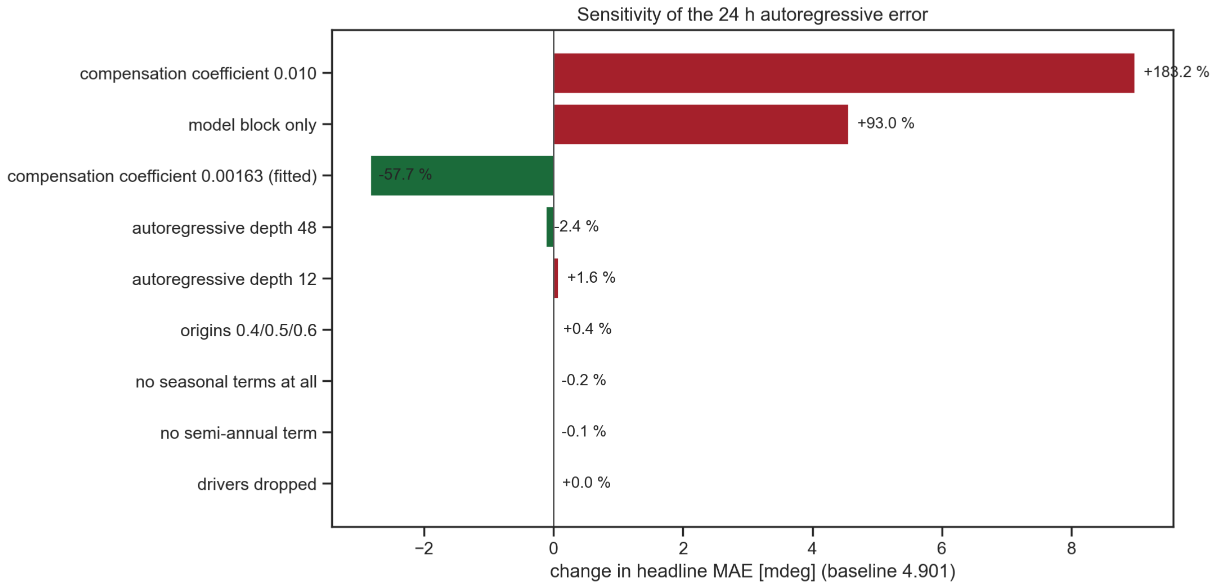

Figure 20: Sensitivity of the 24-hour autoregressive error to the study's choices.

The compensation coefficient is the most consequential decision in the study, moving the headline by −58 % to +183 %. Every absolute error in this report is a statement about a target built with a particular coefficient, and that coefficient has never been calibrated for this instrument.

The seasonal decomposition contributes nothing to prediction. Removing every seasonal term changes the 24-hour error by 0.2 %. Sections 3 and 4 establish that the cycles are real, identified and interpretable; this table and Section 8 establish that they are useless for forecasting, because they are not stationary. Both statements are true and the report keeps them separate.

The drivers are exactly worthless. Dropping them changes the headline by 0.00 % — the third independent route to that conclusion, after the unbroken-block ranking and the out-of-period test.

10 How much of the record can actually be used

Section 2 divided the study on a premise: that the spectral and harmonic work tolerates gaps by construction while the autoregressive work does not, and must therefore run on the longest unbroken block. The first half is right. The second half is too strong, and every result from Section 8 onwards was computed under it.

An autoregressive model does not need a continuous series. It needs windows of n_{lags} consecutive observations followed by $n_{\mathrm{forecasts}}$ more, and those windows exist in every block, not only in the longest one. The restriction was never tested; this section tests it.

That also reframes what imputation is for. Filling a hole is not a way of making a record continuous — it is a way of buying back the windows the hole destroyed. A single missing hour costs a full window's worth of samples however brief it is, so the many short interruptions are cheap to fill and expensive to leave, and a months-long outage is the reverse.

10.1 What each gap tolerance buys

Section 2 swept the tolerance to three days. Extended, it shows where the record actually divides.

tolerance	blocks	longest	cycles	fabricated	two cycles?
6 h	11	211.5 d	0.58	1.6 %	fail
24 h	6	600.2 d	1.64	2.6 %	fail
72 h	6	617.2 d	1.69	3.2 %	fail
7 d	6	617.2 d	1.69	3.2 %	fail
30 d	5	617.2 d	1.69	3.2 %	fail
45 d	3	617.2 d	1.69	3.2 %	fail
60 d	2	**1353.9 d**	**3.71**	**14.3 %**	pass
90 d	2	1353.9 d	3.71	14.3 %	pass

Between three days and forty-five the longest block does not move at all. The record holds exactly seven interruptions longer than three days, and their positions explain the whole table:

From	To	Length
2018-09-23	2018-11-21	59.0 d
2019-01-14	2019-02-04	20.8 d
2019-04-20	2019-05-11	20.5 d
2020-06-04	2020-07-31	57.3 d
2022-04-09	**2023-06-21**	**437.6 d**
2024-05-04	2024-06-14	41.5 d
2024-08-27	2024-10-09	42.7 d

The four in the earlier half stand between the record's start and the 617-day block, so absorbing any subset of them short of all four leaves that block unchanged — which is why the longest span is flat from three days to forty-five. The binding constraint is the **59-day** interruption of autumn 2018, and the table jumps precisely when the tolerance passes it: at **sixty days** the whole earlier half consolidates into one stretch of 1353.9 days — **3.71 annual cycles**, the first window in this project to clear the two-cycle rule — at the price of inventing 14.3 % of it.

The record is permanently in two halves. The 438-day outage between April 2022 and June 2023 never closes at any tolerance a person would defend. The earlier half reaches 3.71 cycles; the later half reaches 1.67 and never qualifies.

10.2 Windows gained against hours invented

Days are not what a model consumes. Counting windows instead, at the configuration Section 8 actually used ($n_{\text{lags}} = 24$, $n_{\text{forecasts}} = 168$):

Policy	cycles	windows	vs block	invented	per hour
model block (control)	1.69	14 641	1.00	492 h	—
full era, no fill	6.57	17 128	1.17	0 h	—
full era, fill 6 h	**6.57**	**33 074**	**2.26**	**881 h**	**20.9**
full era, fill 24 h	6.57	39 345	2.69	1 756 h	14.1
full era, fill 72 h	6.57	39 792	2.72	1 821 h	13.8

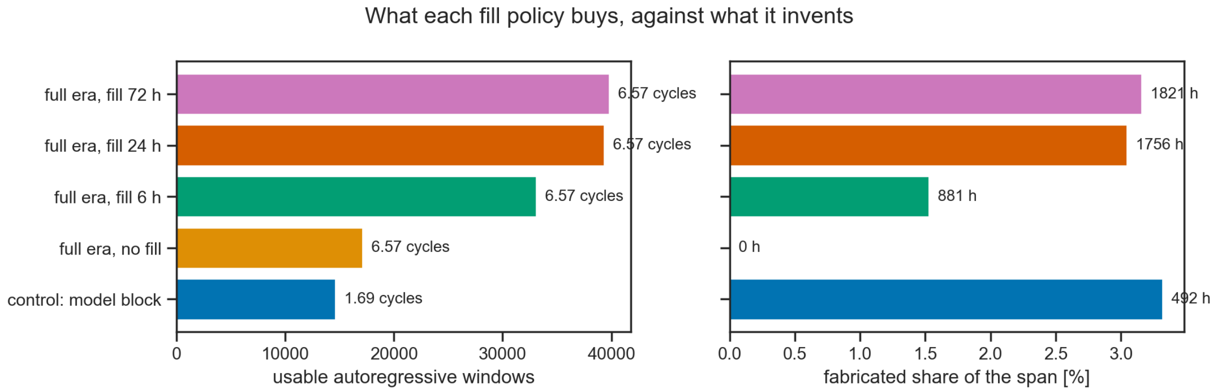

Figure 21: Left: usable autoregressive windows under each policy, annotated with the annual cycles the span contains. Right: the share of that span which had to be invented to obtain them.

Filling the short interruptions is the best-value operation in the study

Taking the whole era *without filling anything* buys only 17 % more windows than the block, because 554 separate interruptions each destroy a full window. Filling just the 475 holes of six hours or less — 881 hours, or **1.5 % of the span**, at a measured interpolation error of 1.1 to 3.2 mdeg against a signal standard deviation of 51.6 — raises that to **2.26 times** the block's windows, and buys **20.9 windows for every hour invented**.

Past six hours the return falls and never recovers: the 24-hour policy buys 14.1 windows per invented hour and the 72-hour policy 13.8. The six-hour limit is also exactly what the Section 5 benchmark licenses for interpolation, so the efficient choice and the defensible one coincide.

10.3 Is this one series, or two?

A 438-day outage is long enough for an instrument to be serviced, remounted or replaced, and an inclinometer's absolute level means nothing across such an event. Fitted straight through it, a trend term would read any offset as structural movement.

The naive comparison confounds the offset with the season, since the last observations before the outage and the first after it sit at opposite ends of the annual cycle.

Test	before	after	step	t
response, raw	151.66	91.55	−60.11	−71.6
response, deseasoned	9.28	22.70	+13.42	+22.9
control `tair`	7.98	24.58	+16.61	+63.5

Response in mdeg, control in °C. Thirty days each side; the deseasoned row has the Section 4 harmonic fit subtracted.

The raw step of −60 mdeg is mostly calendar: the control is 16.6 °C warmer after the outage than before it, and the compensated inclination moves against temperature. Once the seasonal expectation is removed, **13.4 mdeg of step survives** — about a quarter of the signal's standard deviation, and far too large to dismiss.

So the era is treated as two series, not one. That is not a compromise: it is what the prediction regime below does anyway, because it presents each complete stretch to the model separately and normalises each one on its own. An offset introduced during an outage therefore cannot propagate into the fit at all.

10.4 Decomposition: does a longer window help?

The companion manuscript states as a framework rule that a window of one to two annual cycles suffices to fit a yearly Fourier component reliably. This record is the only place in the project able to test it. The same NeuralProphet configuration is fitted on 1.69 cycles and on 6.57, and both are measured against the harmonic fit of Section 4, whose single fixed basis gives an annual signal exactly one place to go.

	model block 1.69 cycles	**full era 6.57 cycles**
observations	14 340	39 499
sd of trend [mdeg]	23.29	35.05
sd of yearly term [mdeg]	35.07	30.07
trend's share	**39.9 %**	**53.8 %**
annual amplitude [mdeg]	48.37	45.48
harmonic reference [mdeg]	42.54	43.31
amplitude ratio	**1.137**	**1.050**
phase gap to reference [d]	10	29

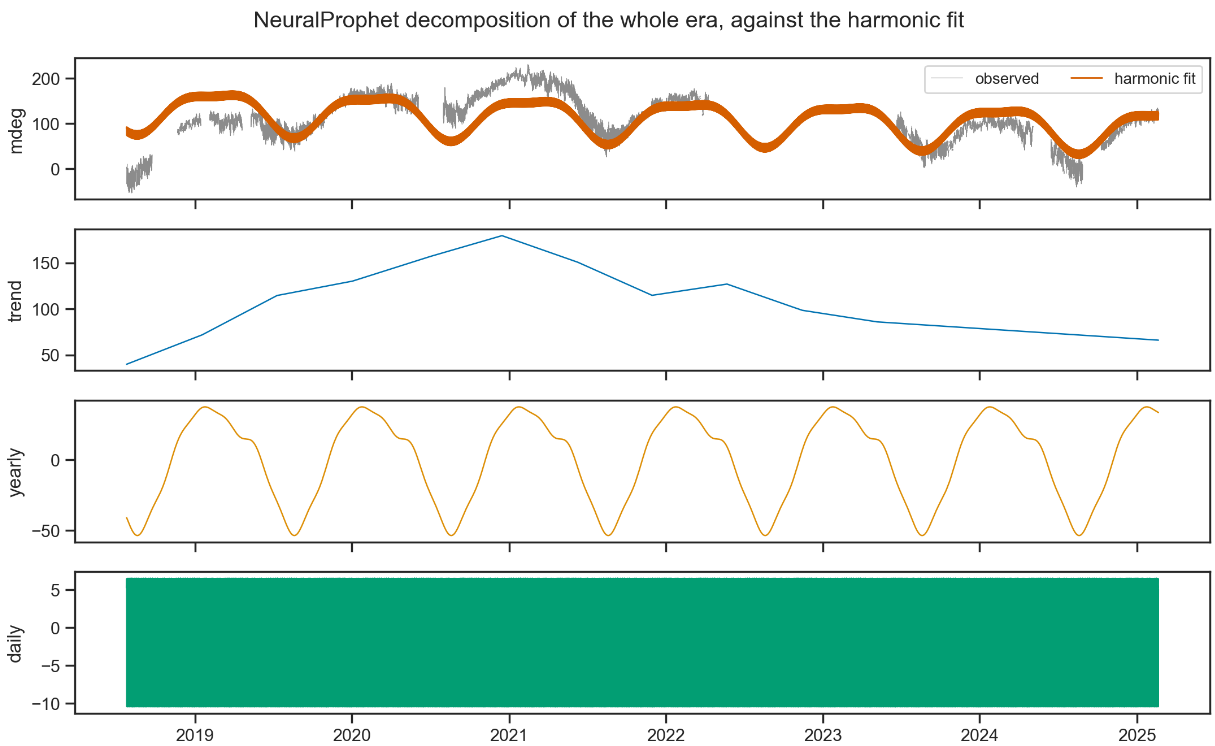

Figure 22: The whole-era decomposition. Top: the observations with the harmonic fit over them; below, the trend and each seasonal component NeuralProphet separated.

More cycles fix the amplitude and make the trend worse

The longer span cuts the annual amplitude's overestimate from **13.7 % to 5.0 %**. On that measure the framework rule is too optimistic: 1.69 cycles identified an annual term nearly fourteen per cent too large, and the extra four cycles largely repaired it.

On every other measure the longer span is no better. The trend's share of the low-frequency variance rises from **39.9 % to 53.8 %** — over the whole era NeuralProphet's piecewise-linear trend holds *more* variance than the annual cycle it is supposed to sit beneath — and the phase drifts from 10 days off the harmonic reference to 29.

The finding is therefore not "use a longer window" but something sharper: **NeuralProphet's trend is not a safe container at either length**. At both spans it absorbs a large share of an annual signal that a fixed harmonic basis attributes correctly, which is the same failure the `thermal_compensation_legacy` study recorded on air temperature and the same term Section 8 priced at 85–93 mdeg when extrapolated. A decomposition that must apportion trend against season correctly should not be read off this component alone.

10.5 Prediction across every complete stretch

NeuralProphet 0.8.0's `drop_missing` does not work

The library's documented mechanism for exactly this situation is `drop_missing`, which is meant to discard the autoregressive windows touching a gap and train on the rest. In version 0.8.0 it does neither correctly. Training on a frame that actually requires a drop returns a **NaN loss from the first epoch** and leaves every weight NaN; the prediction path raises `Length of values ... does not match length of index`, because the rows it drops are not dropped from the frame the predictions are written into. Both were verified on this record.

The route used instead is the library's multi-series support, which is sound: the surviving stretches are presented through the `ID` column, one set of shared components is fitted across all of them, and normalisation is per series. The seasonal terms therefore see all six and a half years while no autoregressive window is ever formed across an outage.

h	block	full era	change	coverage	width
lagged — the deployable configuration					
1	3.35	1.88	−43.8 %	100 → 84	164 → 12
3	4.55	3.19	−29.7 %	100 → 80	170 → 14
6	4.41	4.15	−5.9 %	100 → 79	167 → 16
12	4.76	5.00	+5.1 %	100 → 77	165 → 17
24	5.66	5.53	−2.3 %	100 → 76	167 → 19
72	8.92	8.69	−2.5 %	100 → 65	175 → 23
168	13.53	11.12	−17.8 %	99 → 55	185 → 23
future — regressors known over the horizon					
1	2.01	1.08	−46.1 %	62 → 82	14 → 10
3	2.31	1.72	−25.6 %	61 → 79	15 → 10
6	2.79	1.86	−33.5 %	61 → 79	16 → 10
12	3.15	2.04	−35.1 %	61 → 78	16 → 10
24	3.56	2.11	−40.7 %	61 → 77	17 → 10
72	4.56	2.84	−37.7 %	63 → 76	20 → 12
168	6.27	3.51	−44.0 %	65 → 76	22 → 13

MAE in mdeg; coverage in per cent against a nominal 90; width in mdeg. Each pair reads block → full era.

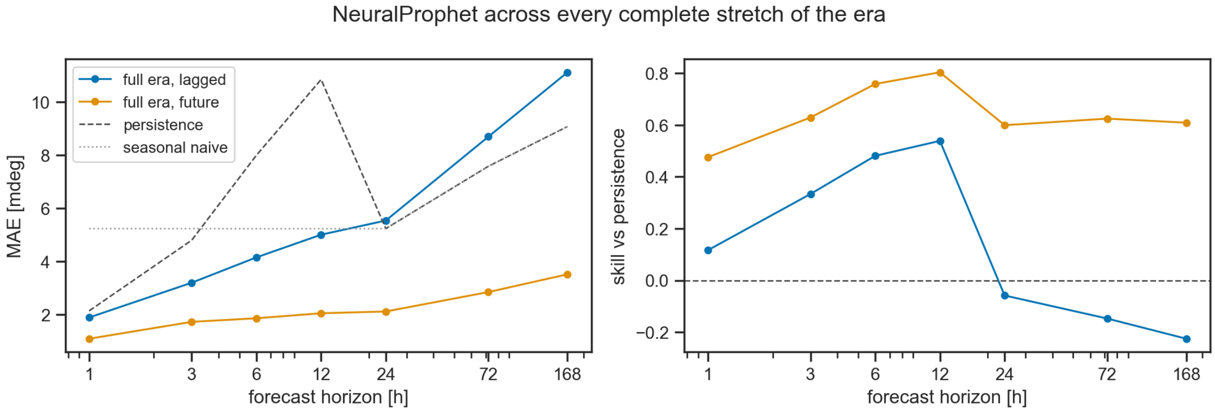

Figure 23: NeuralProphet fitted across every complete stretch of the era, against the persistence and seasonal-naive baselines.

The uncertainty was an artefact of the window, not of the model

Section 8.4 reported that the deployable configuration covered **100 %** of observations with intervals of **164–185 mdeg** — more than three times the signal's own standard deviation, so wide that they excluded nothing and detected nothing. That was the single worst result in this report.

Fitted across the whole record instead, the same configuration produces intervals of **12–23 mdeg** at a coverage of **55–84 %**. The intervals are now narrower than the signal and are no longer trivially satisfied; they under-cover at the long horizons, which a split-conformal recalibration can correct and an interval three times too wide cannot.

The point estimate improves too, by 44 % one hour ahead and 18 % one week ahead, and the useful band against persistence widens from 6–12 h to 1–12 h. With known future regressors every horizon improves, by 26 to 46 %.

None of this comes from a better model. It is the same architecture, the same depth, the same drivers and the same number of epochs. It comes from 2.26 times the training windows over 3.9 times the seasonal span, bought with **1.5 % fabricated data**.

10.6 The bridged block, and what the fill costs it

A genuinely continuous series of two annual cycles can be had only by inventing the interruptions inside it. The block is built here because the question was asked, and it is reported with its cost attached: 1353.9 days at 3.71 cycles, of which **14.3 % is fabricated**, including six holes longer than anything the Section 5 benchmark licenses and one of 59 days.

Fitting the harmonic model twice — once on the bridged series, once with the fabricated positions masked out again — prices the distortion directly.

Component	bridged	observed only	bias
annual	43.41	44.93	−3.4 %
semi-annual	13.17	15.86	**−17.0** %
diurnal	7.16	8.43	**−15.0** %
semi-diurnal	2.49	2.91	**−14.6** %

Amplitudes in mdeg.

The fill suppresses every component, and the short ones most. A straight line drawn across a 59-day hole carries the annual level tolerably — the annual amplitude falls by only 3.4 % — but it contains no diurnal variation at all, so the components with periods shorter than the hole lose 15 to 17 % of their amplitude. The bias is systematically downward, because interpolation adds span without adding oscillation.

This block is not recommended and nothing else here depends on it. Regime B obtains 6.57 annual cycles for 1.5 % fabricated data; this obtains 3.71 for 14.3 %. The bridged block would be worth its cost only if a strictly uninterrupted series were required for some other reason, and no analysis in this report requires one.

11 Verdicts

Question	Answer
1 · periodicities	Annual at 370.1 d (power 0.400), semi-annual at 183.0 d, diurnal at 23.93 h with second and third harmonics, and an unanticipated component at 113.9 d. All clear a 200-replicate permutation null.
1 · spurious peaks	The 971-day feature has more window-function power than signal power: an outage artefact. The 7.28-day feature is an artefact of the detrending filter.
2 · is 180 d independent?	**No.** Phase-locked to the annual term at 28.5° against 103.9° for an unlocked null: the annual second harmonic. Frequency alone could never separate them — that needs 34 years.
2 · best decomposition window	The whole era, necessarily. No contiguous block holds two annual cycles; the longest holds 1.69.
2 · are the components stable?	**fail**. Annual amplitude 30.6–53.3 mdeg, phase spanning 25 days.
3 · imputation limit	Interpolation to 6 h, seasonal-naive to 24 h, model-based fillers never. This addresses 9.6 % of the missing hours; the 84.5 % in gaps over 30 days is not recoverable.
4 · best single predictor	`tair`, at $R^2 = +0.073$ on the whole era. **fail** on the unbroken block, where no driver beats a constant ($R^2 = -0.719$).
4 · lag structure	Both forcings instantaneous. The signed control passes: the only negative optimum is `rh` at -1 h, buying 0.0018 R^2.
4 · negative control	`batt` ranks *second of three* on the whole era, above relative humidity — the noise floor sits above one of the two real channels.
5 · best combination	`tair + rh`, by 1.3 % against a fold spread twenty-three times larger. **fail** as a resolved ordering.
5 · out-of-period	**fail**. Both models roughly double their error, to 62–64 mdeg — worse than the 48.8 mdeg a constant achieves.
6 · error budget	Autoregression is 47.7 of the 48.8 mdeg at 1 h and still dominates at one month. Seasonal terms contribute 0.1 mdeg; drivers contribute nothing.
6 · useful horizon	Autoregression beats climatology out to 4380 h (six months). NeuralProphet's deployable configuration beats persistence only from 6 to 12 h.
6 · the trend term	**fail**. Extrapolating it costs 85–93 mdeg at every horizon.
6 · uncertainty	**fail** both ways *on the unbroken block*: the deployable configuration covers 100 % with intervals over three times the signal's standard deviation; the perfect-foresight one covers 61–65 %. Section 10 largely repairs this.

Question 7 was raised by the answers above and is Section 10.

Question	Answer
7 · must autoregression use one block?	**No**, and the assumption was expensive. A model needs windows, not a continuous series, and they exist in every block.
7 · usable record	The longest block gives 1.69 annual cycles and 14 641 windows. Fitting across every complete stretch gives **6.57 cycles and 33 074 windows** — **2.26 times** as many — for **1.5 %** fabricated data.
7 · where imputation pays	Only below six hours: 475 short gaps cost 881 h and buy **21 windows per invented hour**. At 24 h the return falls to 14.1, and the six-hour limit is also what the benchmark licenses.
7 · one series or two?	**Two.** The 438-day outage shows a −60.1 mdeg step, of which +13.4 mdeg survives removing the seasonal expectation — a quarter of the signal's standard deviation. Each stretch is normalised separately.
7 · does a longer window fix the decomposition?	**Partly.** The annual amplitude's overestimate falls from 13.7 % to 5.0 %, but the trend's share of low-frequency variance *rises* from 39.9 % to 53.8 %. **fail** as a general fix: NeuralProphet's trend absorbs the annual cycle at either length.
7 · prediction, full era vs block	Lagged MAE −44 % at 1 h and −18 % at 168 h; intervals fall from 164–185 mdeg at 100 % coverage to **12–23 mdeg at 55–84 %**. Future regime improves 26–46 % at every horizon.
7 · the bridged block	3.71 cycles for **14.3 %** fabricated data, biasing the annual amplitude −3.4 % and every shorter component by 15–17 %. Reported, not recommended.
7 · `drop_missing`	**fail** — NeuralProphet 0.8.0's own mechanism for gapped autoregression returns a NaN loss from the first epoch and cannot predict at all. The multi-series route is used instead.

12 Caveats

The target is constructed, and the sensitivity table shows how much that matters. Every absolute error depends on a compensation coefficient never calibrated for this instrument. The relative statements — which components matter, which do not — are far more robust than the levels.

The sign contradiction is now two-instrument and still unresolved. Section 1.3 adds evidence but does not settle it.

The 113.9-day component is recorded, not explained. It is genuine by the window-function and permutation controls, and this report offers no mechanism.

Non-stationarity is diagnosed, not modelled. The study establishes that the annual amplitude and phase move; it does not fit a time-varying seasonal model, which is the obvious next step and would change the Section 8 conclusions if it succeeded.

One station, one channel. The legacy network had three blocks. This study reads only `st02` for comparability with the extended-sensor study, so it cannot separate structural from instrument behaviour by cross-station comparison — the test most likely to resolve Section 1.3.

The 438-day outage cannot be bridged, and the bridged block is not a recommendation. Section 10 builds it because the question was asked and because the cost of an answer should be measured rather than asserted. The holes it fills are far longer than anything the Section 5 benchmark licenses, and no result in this report depends on it.

Sections 1 to 6 keep the longest-block restriction. They were computed under it, and they are left that way so that every ported result stays comparable with the extended-sensor study, which had no choice in the matter. Section 10 measures what the restriction costs; it does not retrofit the earlier sections with the answer.

13 Artefact inventory

Artefacts are numbered in reading order.

Artefact	Content
S4_01 – S4_04	Channel summary, coverage, gap inventory, contiguous blocks
S4_05 – S4_07	Sign test; correlation matrices, levels and differences
S4_08, S4_09	Spectral peaks, low and high band
S4_10 – S4_13	Harmonic fit, sliding fit, stability, phase-lock test
S4_14, S4_15	Imputation benchmark and policy
S4_16 – S4_19	Operators full band, diurnal band, signed; phase control
S4_20	Variance inflation factors
S4_21, S4_22	Single-predictor rankings, whole era and unbroken block
S4_23	Exhaustive subset search
S4_24 – S4_26	Leave-one-out gain, permutation importance, out-of-period test
S4_27	The diagnostic protocol
S4_28 – S4_31	Error budget, marginal gains, trend cost, regime gap
S4_32 – S4_34	NeuralProphet horizons, limits, joint head
S4_35, S4_36	Sensitivity sweep, verdicts
S4_37, S4_38	Gap-tolerance sweep; window accounting per fill policy
S4_39, S4_40	Baseline step across the long outage; decomposition, control against the whole era
S4_41 – S4_43	Segmented horizons, gain over the block, bridged-block bias
S4_F01, S4_F02	Overview and target
S4_F03, S4_F04	Spectra, low and high band
S4_F05 – S4_F07	Harmonic fit, sliding harmonics, diurnal cycle
S4_F08, S4_F09	Gaps and imputation; correlations
S4_F10 – S4_F12	Operator grids: full band, diurnal, signed
S4_F13 – S4_F15	Ranking, subsets, contribution diagnostics
S4_F16	Error budget
S4_F17 – S4_F19	NeuralProphet skill, calibration, fan chart
S4_F20	Sensitivity
S4_F21 – S4_F23	Window accounting; whole-era decomposition; segmented skill

Reproduction, from the study directory with `neuralprophet_env` active:

```
jupytext --to ipynb inclination_prediction_legacy_study.py
jupyter nbconvert --to notebook --execute --inplace
                  inclination_prediction_legacy_study.ipynb
```